ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. EE ENGINEERING — SEMESTER 5

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain structure of modern electric power system. Compare AC vs DC transmission systems. [14 Marks]

[OR]

Q.1 (Alt) Derive expressions for Line Inductance and Capacitance of 3-phase transposed overhead transmission line. [14 Marks]

--- UNIT II ---

Q.2 Classify transmission lines (Short, Medium, Long). Derive ABCD parameters using Nominal-T and Nominal-Pi. [14 Marks]

[OR]

Q.2 (Alt) Explain Ferranti Effect and Surge Impedance Loading (SIL) of long transmission lines. [14 Marks]

--- UNIT III ---

Q.3 Explain Overhead Line Insulators (Pin, Suspension, Strain). Derive String Efficiency and methods to improve it. [14 Marks]

[OR]

Q.3 (Alt) Derive Sag and Tension equations of overhead conductor supported at equal and unequal levels. [14 Marks]

--- UNIT IV ---

Q.4 Explain Corona phenomenon in overhead lines: Disruptive critical voltage and power loss formula. [14 Marks]

[OR]

Q.4 (Alt) Describe Underground Cables: Construction, Classification, Insulation resistance, and Grading of cables. [14 Marks]

--- UNIT V ---

Q.5 Explain Distribution Systems: Radial, Ring Main, and Interconnected systems. Kelvin's Law for conductor size. [14 Marks]

[OR]

Q.5 (Alt) Describe Substation grounding systems (solid, resistance, reactance grounding) and equipment layout. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****